SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 3.2: Probability I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.2: Probability I
Driving Question	DRIVING QUESTION: "How can Kamau — and other Kenyans facing uncertain everyday outcomes — use the mathematics of probability to make smarter, fairer, and safer decisions?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring the Phenomenon & Experimental Probability: Why Can't Kamau Predict His Full Bus?**	Students observed that Kamau's bus from Githurai to Nairobi CBD is sometimes full and sometimes not — a real, unpredictable daily outcome. Through a class simulation (e.g., flipping a coin or drawing coloured counters to represent 'full' vs 'not full' buses), students gathered frequency data across multiple trials. They recorded how often the 'full bus' outcome occurred out of total trials, and noticed that results varied between small and large samples. Pedagogical note: ensure students physically record each trial result in a tally table before any calculation — do not shortcut to answers.	Students learned that experimental probability (also called relative frequency) is calculated as: P(event) = (number of times the event occurred) ÷ (total number of trials). They learned that with fewer trials the relative frequency fluctuates widely, but as the number of trials increases it tends to stabilise toward a consistent value — the Law of Large Numbers in action. They also established that probability values always lie between 0 (impossible) and 1 (certain), and can be expressed as fractions, decimals, or percentages. Pedagogical note: explicitly record on the board how the running relative frequency changes after 5, 10, 20, and 50 trials to make the stabilisation pattern visible.	Students explained that Kamau cannot perfectly predict any single bus trip because each outcome is the result of a random process influenced by many uncontrolled factors (time of day, matatu delays, school runs in Githurai). However, by tracking outcomes over many trips, Kamau can calculate an experimental probability that gives him a realistic expectation — e.g., 'In 40 recorded mornings, the bus was full 28 times, so P(full) ≈ 28/40 = 0.70.' This grounds the driving question: probability does not eliminate uncertainty but quantifies it, enabling smarter planning. Pedagogical note: connect back to the anchoring phenomenon explicitly — ask, 'How does this first tool begin to help Kamau make a smarter decision?'
Lesson 2 **Complementary Events and the Probability Scale: The Other Side of Kamau's Coin**	Students observed that every outcome in Kamau's bus scenario has an opposite: if the bus is full, it is NOT available; if it is not full, it IS available. Using their experimental data from Lesson 1, students identified pairs of complementary outcomes and plotted various everyday Kenyan events	Students learned that for any event E, its complement E' (read 'E prime' or 'not E') consists of all outcomes in the sample space that are NOT in E. The Complementary Rule states: P(E) + P(E') = 1, therefore P(E') = 1 − P(E). They learned to apply this rule to calculate a missing	Students explained that if Kamau records P(bus full) = 0.70, then by the Complementary Rule, P(bus NOT full) = 1 − 0.70 = 0.30 — without needing to recount. They applied this to multi-step Kenyan scenarios: e.g., 'A maize farmer in the Rift Valley estimates P(good harvest) = 0.65;

	(e.g., 'it rains in Kisumu today', 'a mandazi is undercooked', 'KCSE results improve') on a 0-to-1 probability scale drawn on the board. Pedagogical note: use a physical washing line strung across the classroom with event cards that students peg at estimated positions — this kinesthetic activity makes the scale concrete.	probability when only one of the pair is known — a highly efficient problem-solving shortcut. They also reinforced the probability scale: impossible (0), unlikely, even chance (0.5), likely, certain (1). Pedagogical note: stress that 'complement' is a precise mathematical term — it means ALL other outcomes, not just one alternative.	therefore P(poor harvest) = 0.35.' Students explained why this rule is powerful for decision-making: Kamau need only track one outcome to know both probabilities, halving his data-collection effort. Pedagogical note: close the lesson by updating the Driving Question Board — students should articulate how complementary events add a second tool to Kamau's decision toolkit.
Lesson 3 **Probability Space (Sample Space): Listing Every Possibility for Kamau's Route**	Students observed a structured scenario: Kamau can board at one of two stops (Githurai 44 or Githurai 45) and the bus can be either a matatu or a full-size bus — creating a two-attribute random experiment. Students used systematic listing methods (ordered lists, two-way tables, and beginnings of tree structures) to enumerate all possible outcomes. They observed that unsystematic guessing led to missed outcomes, while organised methods reliably produced a complete list. Pedagogical note: deliberately allow one group to list unsystematically first, then reveal the missed outcomes — this creates productive cognitive conflict that motivates the need for systematic methods.	Students learned that the sample space (S) — also called the probability space — is the complete set of ALL possible outcomes of a random experiment. They learned three methods for constructing a sample space: (1) systematic listing, (2) a two-way table (grid method), and (3) a preliminary tree diagram. They learned that the theoretical probability of an event E is P(E) = n(E) ÷ n(S), where n(E) is the number of outcomes in E and n(S) is the total number of equally likely outcomes. Pedagogical note: insist students write sample spaces using set notation { } and label each element clearly — this habit becomes essential in Lessons 5–7.	Students explained that by listing the full sample space for Kamau's route choices — S = {(G44, matatu), (G44, full-bus), (G45, matatu), (G45, full-bus)} — they could calculate exact theoretical probabilities for any event of interest, e.g., P(Kamau boards a matatu) = 2/4 = 0.5. They connected this to the driving question: knowing the full probability space means Kamau can see ALL possibilities simultaneously, not just guess from memory. They also explained the difference between experimental probability (Lesson 1) and theoretical probability: theory assumes equally likely outcomes; experiment uses real collected data. Pedagogical note: use a Venn circle on the board to show S as the universal set containing all events discussed.
Lesson 4 **Theoretical Probability and Single Events: What Should Kamau Expect?**	Students observed a carefully designed single-event probability scenario: a bag contains tokens representing different matatu routes Kamau could take from Nairobi — each token labelled with a route (e.g., 23, 44, 58, 111, 125). Students drew tokens at random (with replacement) and recorded outcomes. They then compared their experimental relative frequencies with calculated theoretical probabilities, observing convergence over many trials. Pedagogical note: run at least 30 draws as a class to give the Law of Large Numbers enough trials to show the convergence clearly — project a running frequency table on the board.	Students learned that theoretical probability is defined as P(E) = (number of favourable outcomes) ÷ (total number of equally likely outcomes). They learned that this formula applies only when all outcomes are equally likely — a condition that must always be checked. They practised computing P(E) for a variety of single events including impossible events (P = 0) and certain events (P = 1), and verified that the sum of probabilities of all mutually exclusive outcomes equals 1. They also practised expressing probability as a simplified fraction, a decimal, and a percentage. Pedagogical note: require students to state the sample space before writing any probability — this structural habit prevents the common error of counting outcomes	Students explained that Kamau can use theoretical probability to set baseline expectations before collecting any data — for example, if 3 of 10 equally likely route tokens represent fast express routes, P(fast route) = 3/10 = 0.30 = 30%. They explained the practical boundary of the formula: it works well for fair, symmetric situations (fair coins, well-shuffled cards, equally loaded bags) but real-world events like weather in Kericho or bus punctuality require experimental data instead. Students articulated this distinction clearly in written explanations, connecting back to Lesson 1's experimental approach and explaining when each method is more appropriate for helping Kamau make decisions. Pedagogical note: cold-call at least 4

		incorrectly.	students to explain the equally-likely-outcomes condition in their own words before moving on.
Lesson 5 **Mutually Exclusive Events and the Addition Law**	Students observed two scenarios side by side: (A) Kamau draws one card from a shuffled pack — can it be both an Ace AND a King in a single draw? (B) Kamau draws one card — can it be both an Ace AND a Heart? Students recorded outcomes in two-way tables and Venn diagrams, observing that in scenario A the two events share zero outcomes, while in scenario B they share one outcome (Ace of Hearts). This visual contrast made the definition of mutual exclusivity tangible. Pedagogical note: use large physical Venn diagram hoops on the classroom floor — place event-outcome cards inside them and let students physically move cards to see overlap or no overlap.	Students learned that two events are mutually exclusive (disjoint) when they cannot both occur in a single trial — their intersection is the empty set ($A \cap B = \varnothing$). They learned the Addition Law for mutually exclusive events: P(A or B) = P(A) + P(B). They also learned the General Addition Law for non-mutually-exclusive events: P(A or B) = P(A) + P(B) − P(A and B), understanding that subtracting the intersection prevents double-counting. They applied both forms to Kenyan contexts: e.g., probability that a randomly selected student from a Nairobi school studies either History or Geography (mutually exclusive timetable slots) versus studies either Kiswahili or is female (non-exclusive). Pedagogical note: use colour-coded Venn diagrams consistently — same colours in every example so the visual language becomes automatic.	Students explained that for Kamau, mutually exclusive thinking helps identify when adding probabilities directly is valid and when it is not. Example: if P(matatu is route 23) = 0.20 and P(matatu is route 44) = 0.35, and a bus can only be one route, these are mutually exclusive — P(route 23 or route 44) = 0.55. They explained the danger of ignoring overlap in non-exclusive events and showed algebraically why the intersection must be subtracted. Students connected this to fairness in decision-making: the Addition Law ensures Kamau never overestimates or underestimates the probability of combined outcomes when planning alternative routes. Pedagogical note: require students to always start by asking 'Can these two events happen at the same time?' before selecting which addition formula to apply.
Lesson 6 **Independent Events and the Multiplication Law: Can Kamau Count on Both Trips?**	Students observed a two-stage experiment: first, a coin flip to represent whether Kamau's morning bus is on time (H = on time, T = late); second, a spinner representing whether his evening bus is on time. Students recorded both outcomes for 20 paired trials and examined whether the morning result appeared to affect the evening result. They also examined a contrasting non-independent scenario: drawing two counters from a bag WITHOUT replacement, where the first draw changes what is available for the second. Pedagogical note: run both experiments in parallel at different table groups and compare class results — the contrast between independent and dependent data sets makes the concept far more memorable.	Students learned the formal definition of independent events: two events A and B are independent if and only if the occurrence of A does not change the probability of B (and vice versa). They learned the Multiplication Law for independent events: P(A and B) = P(A) × P(B). They learned to test independence by checking whether P(A and B) = P(A) × P(B) holds in a given data set. They also learned to extend the multiplication law to three or more independent events: P(A and B and C) = P(A) × P(B) × P(C). Pedagogical note: explicitly contrast this lesson's multiplication with the addition from Lesson 5 — draw a decision flowchart on the board: 'OR → Add; AND (independent) → Multiply.'	Students explained that Kamau's morning and evening bus outcomes are independent — missing the morning bus does not make the evening bus more or less likely to be on time. Therefore, if P(morning on time) = 0.75 and P(evening on time) = 0.80, then P(both on time) = 0.75 × 0.80 = 0.60. They applied this to other Kenyan independent scenarios: e.g., P(Kipchoge wins a race) and P(it rains in Mombasa today) are independent events. They also explained why the equally-likely assumption is not required here — the multiplication law works for any probabilities as long as independence holds. Pedagogical note: close with a misconception check — ask students to identify and correct the error in: 'P(A) = 0.4, P(B) = 0.5, so P(A and B) = 0.9.'
Lesson 7 **Probability Tree Diagrams: Mapping All**	Students observed a fully worked tree diagram for Kamau's two-trip day: Branch 1 (morning trip): on time (0.75) / late (0.25);	Students learned that: (1) a tree diagram displays ALL possible outcomes of sequential events simultaneously in a single	Students explained that the tree diagram is Kamau's most powerful tool so far because it integrates all prior lessons into one visual

of Kamau's Possible Outcomes	Branch 2 (evening trip, drawn from each morning branch): on time (0.80) / late (0.20). Students traced each complete path (branch-end) from root to leaf, listed the four combined outcomes, and observed that multiplying along branches and adding across final branches each served a distinct and necessary purpose. They also observed that the probabilities at all final branches must sum to 1 — a built-in self-checking mechanism. Pedagogical note: have students draw the tree on large A3 paper first by hand before any printed version — the physical drawing process builds structural understanding that a pre-drawn diagram cannot replicate.	organised structure; (2) probabilities are MULTIPLIED along each branch path to find the probability of that complete sequence (applying the Multiplication Law from Lesson 6); (3) probabilities of mutually exclusive final outcomes are ADDED to find the probability of a combined event (applying the Addition Law from Lesson 5); (4) all end-branch probabilities must sum to 1 — this is a mandatory verification step. They also learned to construct tree diagrams for both equally and unequally probable branches, and for two-stage and three-stage experiments. Pedagogical note: explicitly label each branch with its probability and each end-node with its combined outcome label AND combined probability — incomplete labels are a major source of student errors.	structure. Using the Lesson 7 tree: P(both on time) = 0.75 × 0.80 = 0.60; P(morning on time, evening late) = 0.75 × 0.20 = 0.15; P(morning late, evening on time) = 0.25 × 0.80 = 0.20; P(both late) = 0.25 × 0.20 = 0.05. Total = 1.00 ✓. Students explained that Kamau can now read off the probability of any combined scenario directly, making decisions like 'Should I carry packed githeri in case I miss the evening bus?' quantitatively informed. Pedagogical note: link every end-branch outcome explicitly back to the driving question — ask, 'Which branch should most concern Kamau, and what decision does its probability suggest he should make?'
Lesson 8 **Consolidation, Real-World Application & Driving Question Resolution: How Has Kamau's Thinking Changed?**	Students observed a capstone case study presenting Kamau's full week of transport data alongside a new Kenyan decision scenario (e.g., a Kisumu fisherwoman estimating the probability of a good catch on two consecutive days, or a Kericho tea farmer assessing the likelihood of two consecutive days with sufficient rainfall). Students reviewed all probability tools introduced across the strand — experimental probability, complementary events, sample space, theoretical probability, mutually exclusive events, the Addition Law, independent events, the Multiplication Law, and tree diagrams — identifying which tool best fitted each part of the scenario. Pedagogical note: present the case study as a genuine problem to solve, not a worked example — groups should debate which tools to apply before committing to calculations.	Students consolidated all eight lessons' key content: (1) Experimental probability = frequency ÷ trials, stabilises with large n. (2) Complementary Rule: P(E') = 1 − P(E). (3) Sample space: complete set of all possible outcomes, written in set notation. (4) Theoretical probability: P(E) = n(E) ÷ n(S) for equally likely outcomes. (5) Mutually exclusive events: A ∩ B = ∅; P(A or B) = P(A) + P(B). (6) General Addition Law: P(A or B) = P(A) + P(B) − P(A ∩ B). (7) Independent events: P(A and B) = P(A) × P(B). (8) Tree diagrams: multiply along branches, add across branches, verify sum = 1. Students also consolidated the meta-skill of selecting the appropriate probability tool for a given real-world context. Pedagogical note: create a visual 'Probability Toolkit' summary chart — either student-generated on the board or a co-constructed poster — that maps each tool to the type of question it answers.	Students explained their final, complete answer to the driving question: Kamau — and all Kenyans facing uncertain outcomes — can use probability mathematics to (a) quantify uncertainty from past data using experimental probability; (b) compute what is theoretically expected under fair or known conditions; (c) combine event probabilities correctly using the Addition and Multiplication Laws depending on whether events are mutually exclusive or independent; and (d) map all sequential possibilities using tree diagrams to evaluate the full range of outcomes before making a decision. Students articulated that probability does not give certainty — it gives calibrated expectation, enabling smarter planning (Kamau leaves earlier when P(late) is high), fairer assessment (not assuming the worst or best without data), and safer choices (a farmer, fisherwoman, or matatu operator makes resource decisions proportional to actual risk). Pedagogical note: close with student-led presentations of Driving Question Board responses — each group should present one real Kenyan decision scenario and demonstrate which probability tool they

			would use and why, completing the full learning arc of the sub-strand.